Retrieval-Augmented Feature Generation for Domain-Specific Classification

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***Abstract*—Feature generation can significantly enhance learning outcomes, particularly for tasks with limited data. An effective way to improve feature generation is to expand the current feature space using existing features and enriching the informational content. However, generating new, interpretable features usually requires domain-specific knowledge on top of the existing features. In this paper, we introduce a Retrieval-Augmented Feature Generation method, RAFG, to generate useful and explainable features specific to domain classification tasks. To increase the interpretability of the generated features, we conduct knowledge retrieval among the existing features in the domain to identify potential feature associations. These associations are expected to help generate useful features. Moreover, we develop a framework based on large language models (LLMs) for feature generation with reasoning to verify the quality of the features during their generation process. Experiments across several datasets in medical, economic, and geographic domains show that our RAFG method can produce high-quality, meaningful features and significantly improve classification performance compared with baseline methods.**

***Index Terms*—feature generation, knowledge retrieval, large language models.**

I. Introduction

In domain-specific applications, e.g., disease classification and insurance claim prediction, the data are usually scarce and result in limited features [1–3]. In these cases, it is infeasible to train a machine learning model with satisfactory performance. A common practice is to generate new features to support the model training. However, manual feature generation relies heavily on domain expertise, while automated methods might produce features that lack interpretability [4, 5]. Existing studies typically focus solely on optimizing data structures [6]. Although efficient, these approaches lack transparency and interpretability. In addition, obtaining the resources of domain-specific knowledge is expensive [7, 8].

An intuitive solution is to utilize the knowledge among existing features and their associations by extracting and understanding their underlying relationships. This process is non-trivial. It requires referencing domain knowledge to generate new features, a challenging task for the models without specific fine-tuning. Although fine-tuning a language model with domain-specific data can inject domain knowledge, it is costly and might limit a model's generalizability across different domains [9, 10]. Thus, an alternative is to retrieve useful information from an external knowledge base as supporting references. Then, a generator model with a reasoning mechanism, e.g., a Large Language Model (LLM), can be used to integrate the original feature with the corresponding support knowledge to generate the new feature. This is similar to the retrieval-augmented generation (RAG) [11–13] technique, where we inherit a similar principle for domain-specific feature generation. On the one hand, the retrieved knowledge is expected to serve as explicit references for interpretability [14, 15]. On the other hand, the new features are generated based on the existing features, which can reduce the representation inconsistent risk and leverage limited data efficiently [16, 17]. The crucial question is *how to incorporate the external retrieved information with the existing feature to support useful feature generation?*

We depict a comprehensive example in Fig. 1 for the feature generation task. The goal is to detect heart disease based on the generated features. To solve the task, a more concrete indicator is the *BMI* feature, which could be generated by the original features *Height* and *Weight*. This desirable information can be generated with the support of external knowledge of *BMI* via retrieval. Such insights serve as our motivation for implementing retrieval-augmented feature generation.

Our Targets. We aim to address three main challenges in generating features for domain-specific scenarios: 1) how to extract and utilize useful information by understanding their relations, 2) how to apply the retrieval-augmented generation paradigm to retrieve useful knowledge with LLMs' reasoning capability for generating informative and explainable domain features, and 3) how to form a high quality feature space on top of the original and generated features.

Our Approach. To address these challenges, we propose an LLM-based **R**etrieval-**A**ugmented **F**eature **G**eneration approach (RAFG) to generate features with retrieved knowledge in terms of the existing features. Specifically, we first instruct an LLM to analyze and capture the implicit pattern within the structured data and the textual information, e.g., dataset descriptions and feature labels. With advanced reasoning capabilities, the LLM is expected to provide a logical process to derive the relation among the features and generate a query for later domain-specific retrieval. Then, we deploy

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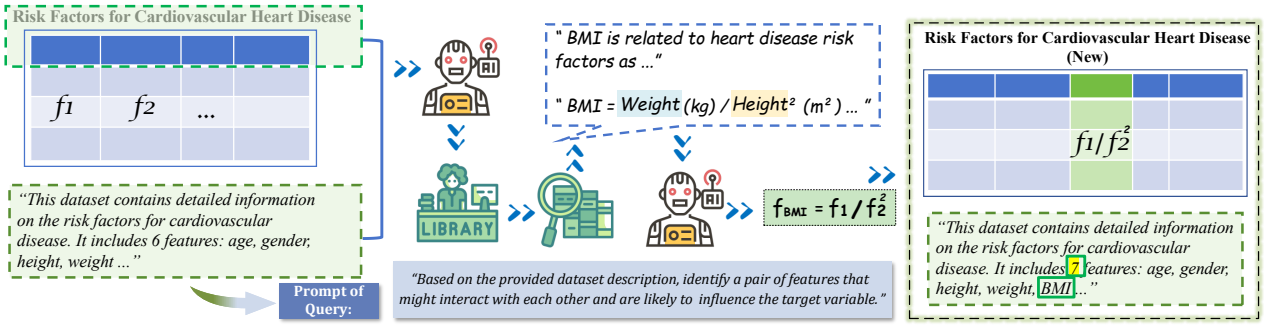


Fig. 1: Framework of RAFG. We adopt an LLM to generate new features according to the retrieved textual information containing expertise knowledge (e.g., the BMI as shown case).

the RAG technique to identify the useful references from a reliable knowledge base according to the previously generated query. The used original features, together with the retrieved knowledge, are fed to the LLM to generate new features according to the reasoning capacity of LLM. Eventually, an automated feature generation mechanism that consistently refines the generated feature and is integrated into the final data attribution. This is supported by an iterative feature validation mechanism to update the feature space with validated ones. The iteratively validated features via generation are expected to enhance the domain-specific interpretability by aligning with the needs of the fields and supply additional crucial indicators.

Our contributions are threefold:

1) We introduce a novel LLM-based retrieval-augmented feature generation method for domain-specific scenarios, which extracts and utilizes the potential relations among the existing features for new feature generation.
2) The RAFG framework is training-free and can be adapted for any specific domains with an automatic validation mechanism. The generated feature is more explainable and enhances the domain-specific classification performance with the retrieved reference.
3) Experimental results on domain-specific downstream tasks demonstrate the effectiveness of our RAFG by outperforming other feature extraction approaches.

II. Related Work

A. Feature Generation

Feature engineering is an essential part of machine learning, including feature selection, feature editing, or generating new features from raw data to improve the model performance [18–20]. The target of feature engineering is to optimize the feature representation space to benefit the downstream tasks [21–23]. Feature generation is an effective approach in the context of feature engineering, which aims to create complex latent feature spaces [24] through various mathematical or logical transformation operations [6, 25, 26]. The advancement of deep learning techniques and LLMs provides new opportunities to enhance the transparency and alleviate the manual operations [27, 28] for feature generation. Existing studies have proposed various methods, ranging from traditional techniques [29] to more sophisticated ones. For instance, Cognito [30] is a reinforcement learning approach for intelligent feature space exploration [31]. Besides, graph-based feature construction [32], policy networks for pre-training feature engineering [33], and multi-stage approaches combining statistical selection with neural generation [34, 35] are developed. Though showing efficiency, these methods still focus on structural and numerical data, which might overlook semantic information and lack interpretability. More recently, CAAFE [36] leverages LLMs to generate context-aware features with explanations [37–41] to enhance the interpretability. Different from them, we propose a retrieval-augmented feature generation framework to facilitate the model's prediction in the downstream tasks with better performance and interpretability.

B. Retrieval-Augmented Machine Learning

Retrieval-Augmented Generation (RAG) [11] aims to retrieve external knowledge to enhance the capacity of language models. The retrieved knowledge is used according to the requirements of downstream tasks [42], such as reducing hallucination [43, 44], supplying up-to-date knowledge for accurate answers [45–47], and assisting in developing better machine learning models via generating useful features [48, 49]. However, the retrieved knowledge could not be directly used for feature generation without further understanding and reasoning by LLMs [32, 38, 50]. To generate interpretable features, the LLM needs to analyze the relevance and rationality of the retrieved information and the generated features [51–53]. Then, the reasoning process is expected to enhance the new features by validating their effectiveness and explainability in the domain knowledge, which is the focus of our studies.

III. Problem Statement

We formulate the task as generating new optimal and explainable feature representations via searching external knowledge on top of the existing feature. Concretely, we denote the original tabular dataset as $\mathcal{D}_0 = \{\mathcal{F}_0; y\}$ that includes an original feature set $\mathcal{F} = \{f_i\}_{i=1}^{I_0}$, each features $\{f_i\}$ and its target label y as well as textual information $\mathcal{C}_0$ including the labels and the data description. Our optimization objective is to

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automatically generate a set of new features $\{g_t\}$ based on the retrieved knowledge with the reasoning procedure supported by an LLM that can reconstruct an optimal feature set $\mathcal{F}^*$ as

$$\mathcal{F}^* = \underset{\hat{\mathcal{F}}}{\text{argmax}}\ \mathcal{P}_{\mathcal{A}}(\hat{\mathcal{F}}, y), \tag{1}$$

where $\mathcal{A}$ is a domain-specific downstream task (e.g., predicting Parkinson's disease), $\mathcal{P}$ is the performance indicator of $\mathcal{A}$ and $\hat{\mathcal{F}} = \{\mathcal{F}_0, \{g_t\}\}$ is an optimized feature set.

IV. Methodology

In this section, we introduce our novel feature generation method **R**etrieval-**A**ugmented **F**eature **G**eneration (**RAFG**), which generates features automatically by the retrieved external knowledge and the original text information. We utilize an LLM as a text-miner and a feature generator, which generates the query for retrieval and integrates the search results with its parametric for feature generation. An automatic mechanism is applied to validate the generated features and the original ones in the feature set, enabling continuous optimization and updating of the feature content. An overview of the **RAFG** framework is shown in Fig. 2, including three stages: *A. Query Generation and Domain Knowledge Retrieval; B. Retrieval-Augmented Feature Generation with Reasoning*; and *C. Feature Content Validation by LLM*.

A. Query Generation and Domain Knowledge Retrieval

We aim to retrieve the external knowledge on top of the original features to augment the information for new feature generation interactively. To this end, given the t-th iteration, we first collect the textual information of the current feature set $\mathcal{F} = \mathcal{F}_{t-1} = \{f_i\}_{i=1}^{I}$, including the goals of the downstream tasks $\mathcal{A}'$, and the textual information $\mathcal{C} = \mathcal{C}_{t-1}$ to construct a query $\mathcal{Q}_t$ via an LLM p_ϕ as

$$\mathcal{Q}_t = p_\phi(\mathcal{A}', \mathcal{C}_{t-1}). \tag{2}$$

We then retrieve the top-k relevant documents from the whole external knowledge base in terms of the query $\mathcal{Q}_t$. The relevance between each candidate document and the query is determined by a computation of cosine similarity as

$$\text{sim}(\mathcal{Q}_t, r_j) = \frac{\mathcal{Q}_t \cdot r_j}{\|\mathcal{Q}_t\| \|r_j\|}, \tag{3}$$

where r_j is the candidate document and $r_j \in \mathcal{R}_k$. Then, the set of retrieved documents $\{\mathcal{R}_k\}$ are considered to consist of a set of potential features $\{g_k\}$ further determined by LLM to update the previous feature set $\mathcal{F}_{t-1}$ as

$$g_k = \{p_\phi(o(\mathcal{F}_{t-1}) \mid \mathcal{R}_k)\}, \tag{4}$$

where o denotes the various operations based on LLM, which are denoted as follows:

- *Scaling*: This operation involves arithmetic transformations, statistical measures, or algorithmic functions applied to derive g_t, i.e., a mapping function $o_s : \mathbb{R}^n \to \mathbb{R}^n$ applied to an existing feature vector to transform into a new one via scaling.
- *Transformation*: This operation involves transforming two or more features into a new feature using concatenation, averaging, or more complex fusion techniques, i.e., applying a mapping function on several existing feature vectors via $o_t : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^p$, where n, m, and p are dimensions of the input features and the resulting feature, respectively.
- *Judgment*: This operation involves logical or rule-based decision-making processes that integrate domain knowledge or empirical rules to form a new feature. In our case, we apply a decision function $o_d : \mathcal{X} \to \{0, 1\}$, where $\mathcal{X}$ is the input space composed of one or more feature vectors, and the output is a new categorical feature based on predefined criteria.

B. Retrieval-Augmented Feature Generation with Reasoning

In this stage, our objective is to generate a new feature and align it with the existing features in the data table. So far, we have obtained external knowledge in terms of the available original features through RAG. The retrieved knowledge contains more structural details and relationships between features, which are not directly reflected in the original dataset. To this end, we adopt the LLM to analyze and extract the potential feature structures and relationships, and further generate new features to expand the feature space.

Given the k documents with potential features $\{g_k\}$, the LLM determines their potential to enhance the performance of downstream tasks via the reasoning capability. To enable the model to conduct high-quality judgments, we employ a "Chain of Thought" [54–56] strategy to encourage the LLM to think and formulate logical hypotheses and reasoning for the possible outcomes of integrating each potential feature into the data table step-by-step. The LLM then selects the most useful documents $\mathcal{R}_t \in \{\mathcal{R}_k\}$ from the candidates, and generates a new feature g_t according to $\mathcal{R}_t$. Finally, we integrate g_t into the data table as

$$g_t = \{p_\phi(o(\mathcal{F}_{t-1}) \mid \mathcal{R}_t)\}, \mathcal{F}_t = \{\mathcal{F}_{t-1}, g_t\}, \tag{5}$$

$$\mathcal{D}_t = \{\mathcal{F}_t; y\} = \{\{\mathcal{F}_{t-1}, g_t\}; y\}. \tag{6}$$

The g_t is generated by the corresponding combination or judgment method o_t mentioned in document $\mathcal{R}_t$ with existing features and matches the length of f_i. Finally, we feed $\mathcal{D}_t$ into the downstream task and obtain the performance metric $\mathcal{P}_t$, which is fed to the LLM for evaluation.

C. Feature Content Validation by LLM

In this stage, our objective is to validate the effectiveness of the newly generated feature. We decide whether to keep the feature by the improvement of $\mathcal{P}_t - \mathcal{P}_{t-1}$, where a higher value of $\mathcal{P}$ indicates better performance as

$$\mathcal{F}_t = \begin{cases} \mathcal{F}_t & \text{if } \mathcal{P}_t > \mathcal{P}_{t-1}, \\ \mathcal{F}_{t-1} & \text{otherwise.} \end{cases} \tag{7}$$

If adding the new feature g_t improves downstream task performance, we formally adopt g_t as a new feature within

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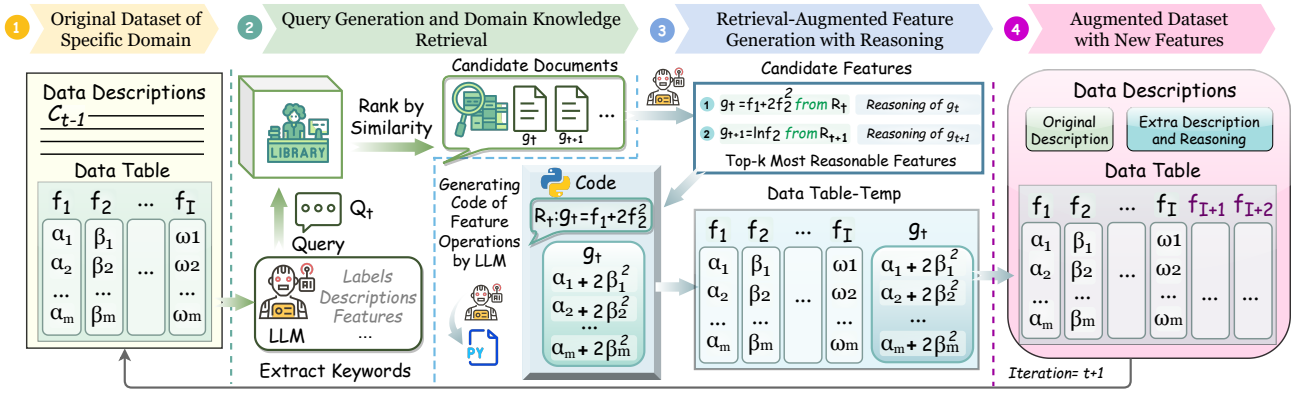


Fig. 2: Framework of RAFG. Given an input data table including a description, feature vectors, and a target label vector, the LLM first integrates the text information of description, label information, and data types to embed and form a query. Then, with this query we adopt RAG technology to search through an external library for one of several relevant documents which can guide the LLM in creating a new feature with most potential. After that, we test the template data table with the new feature for metrics improvement, and the LLM decides whether to reserve this new feature. This searching and generation process iterates until reaching the maximum rounds of iteration, or the best feature space is found.

the table. The LLM updates the dataset text information $\mathcal{C} = \mathcal{C}_t = p_\phi(\mathcal{C}_{t-1}|\mathcal{R}_t)$ to incorporate the information related to g_t. The feature generation process increases the feature space's dimensionality. As the feature space expands, the model can understand and interpret the data more comprehensively, further improving downstream task performance and enhancing generalization capabilities.

We generate new queries to search for potential features with the updated data table $\mathcal{D}_t$ and the data description $\mathcal{C}_t$. This iteration process continues until a predefined maximum of iterations T is reached, or the optimal feature set is found when achieving peak performance in the downstream task. Algorithm 1 presents the pseudocode of the proposed method.

V. Experiments

In this section, we present a series of experiments to evaluate the effectiveness and impacts of the RAFG. We begin by comparing the performance of the RAFG against several baseline methods on four downstream tasks. To further understand the contributions of different components, we then conduct ablation studies investigating the impact of using an external knowledge library and the choice of different LLMs. Subsequently, we present the information gain observed during the feature generation process. We then showcase the relationship between the newly generated features and the original features, followed by a discussion on the reasons for the observed performance improvements.

A. Experimental Settings

Datasets. We evaluate the RAFG on four real-world datasets, including *Parkinson's Disease Classification (PDC)* [57], *Animal Information Dataset (AID)* [58], *Global Country Information Dataset 2023 (GCI)* [59] and *Diabetes Health Indicators Dataset (DIA)* [60]. Table I shows the detailed information.

TABLE I: Datasets from four specific different domains.

Datasets	Samples	Features	Class	Target
PDC	756	754	2	Parkinson Binary
AID	205	15	10	Social Structure
GCI	195	34	4	Life Expectancy
DIA	253680	21	2	Diabetes Binary

Metrics. We evaluate the model performance by the following metrics: *Overall Accuracy (Acc)* measures the proportion of true results in the total dataset. *Precision (Prec)* reflects the ratio of true positive predictions to all positive predictions for each class. *Recall (Rec)*, also known as sensitivity, reflects the ratio of true positive predictions to all actual positives for each class. *F-Measure (F1)* is the harmonic mean of precision and recall, providing a single score that balances both metrics. We calculate Precision, Recall, and F1-score via macro averaging.

Aside from evaluating model performance, to further assess the contribution of the features generated by RAFG, we also utilize *Information Gain (IG)* [61]. This metric quantifies the amount of additional information about the target variable Y that the new features introduce into the dataset. We adopt the concept of information entropy [62] to calculate IG:

$$H(Y|F) = -\sum_{y \in Y} p(y|F) \log p(y|F), \tag{8}$$

where $p(y|F)$ is the probability of class y given the feature set F. The information gain from expanding an initial feature set F_0 to a new feature set F is then computed as:

$$IG(Y; F, F_0) = H(Y|F_0) - H(Y|F), \tag{9}$$

which reflects the informational enrichment provided by the generated features [63]. Prior studies indicate that features yielding higher information gain are valuable for enhancing predictive model performance [64].

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Algorithm 1 Retrieval-Augmented Feature Generation

Input: Tabular dataset $\mathcal{D}_0 = \{\mathcal{F}_0; y\}$, textual information $\mathcal{C}_0$, downstream task model $\mathcal{A}$, maximum iterations T, patience K, LLM p_ϕ, performance metric $\mathcal{P}$

Output: Enhanced feature set $\mathcal{F}$

Initialization:

$\mathcal{F} \leftarrow \mathcal{F}_0$, $\mathcal{C} \leftarrow \mathcal{C}_0$, $\mathcal{D} \leftarrow \mathcal{D}_0$, no_improve $\leftarrow 0$;
$\mathcal{P} \leftarrow \mathcal{A}(\mathcal{D})$; `// Base performance on original features`

for $t = 1$ **to** T **do**
- `// --- Query Generation and Knowledge Retrieval ---`
- $Q_t \leftarrow p_\phi(\mathcal{D}, \mathcal{C})$ Retrieve top-k documents $\{R_k\}$ from knowledge base
- `// --- Candidate Evaluation ---`
- **foreach** R *in* $\{R_k\}$ **do**
  - $o_k \leftarrow p_\phi(R, \mathcal{F}, \mathcal{C})$ `// LLM performs deterministic operation for feature`
  - $g_k \leftarrow o_k(\mathcal{F}_{\text{train}})$ $\mathcal{D}_{\text{tmp}} \leftarrow \{\{\mathcal{F}, g_k\}; y\}$ $\mathcal{P}_k \leftarrow \mathcal{A}(\mathcal{D}_{\text{tmp}})$
- **end**
- Select $R_t = \arg\max_k \mathcal{P}_k$ and its corresponding g_t
- `// --- Validation and Feature Adoption ---`
- **if** $\mathcal{P}_t > \mathcal{P}_{t-1}$ **then**
  - $\mathcal{P} \leftarrow \mathcal{P}_t$; $\mathcal{F} \leftarrow \{\mathcal{F}, g_t\}$ $\mathcal{C} \leftarrow p_\phi(\mathcal{C} \mid R_t)$ `// Update description when feature accepted`
  - no_improve $\leftarrow 0$
- **end**
- **else**
  - no_improve $\leftarrow$ no_improve $+ 1$
- **end**
- **if** *no_improve* $\geq K$ **then**
  - **break** `// Early stopping when no improvement for` K `rounds`
- **end**

end

return $\mathcal{F}$

Baseline Methods. We adapt the RAFG across a range of classification models, including *Random Forests (RF)* [65], *Decision Tree (DT)* [66], *TabTransformer (TT)* [67] and *TabNet (TN)* [68]. We compare the performance of each model with and without the features generated by RAFG. Our RAFG is also compared against several baselines: the raw data (Raw), the Least Absolute Shrinkage and Selection Operator (Lasso) [29], the Feature Engineering for Predictive Modeling using Reinforcement Learning [31] (RL), and the Context-Aware Automated Feature Engineering (CAAFE) [36].

Implementation Details. Experiments are conducted on an NVIDIA RTX 4090 GPU. In our implementation for feature generation, we use GPT-4o [69] as the query generator paired with Google's Custom Search API to perform the retrieval tasks. We use the top-3 results as the retrieved knowledge. GPT-4o then uses these top relevant results for feature generation. For the compared baselines, Lasso and RL are implemented based on standard libraries (scikit-learn [70] and transformers [71]) with hyperparameters tuned via 5-fold cross-validation. CAAFE is configured with default setting with GPT-4o as the generator, and sets the iterations to 3.

B. Experimental Results

Overall Performance. Table II shows the overall results across four domain-specific datasets. All overall performance results are based on the average of 5-fold cross-validation. Compared with baseline methods, RAFG consistently achieves superior performance. For instance, in RF, RAFG shows a significant improvement in accuracy across all datasets, with a 19.2% increase on AID over raw data, and 3.8% over CAAFE, the second-highest accuracy baseline. The RAFG shows particular strength in boosting model accuracy. The performance demonstrates that RAFG's enriching of feature space with domain-specific knowledge enhances the model's understanding and boosts overall performance.

Across different models and metrics, RAFG consistently outperforms other methods in all metrics. Notably, in GCI dataset under DT, RAFG achieves a precision of 66.9%, surpassing the highest precision of 60.7%. Moreover, for F1 scores, RAFG raises the score in the GCI under DT from 55.6% to 68.1%. These results demonstrate the RAFG's capability of reducing misclassification effectively.

The generation of new features through RAFG significantly enhances model performance. As results show, adding features, e.g., from 756 to 770 in the PDC dataset, leads to substantial increases in performance metrics, including accuracy and F1. This result suggests that the new features not only numerically expand the feature set, but also enhance the information richness of the dataset. Thus, models can capture more complex patterns and relationships within the data.

We further assessed the robustness of RAFG across diverse LLMs: GPT-4o [69], Llama 2-13B [72], and Mistral-7B [73]. As shown in Fig. 3, integrating RAFG consistently improves the classification accuracy of the RF classifier on all datasets compared with the Raw baseline. For example, on the PDC dataset, accuracy rises from 0.820 (Raw) to 0.888 with GPT-4o, and similar gains are observed with Llama 2-13B (0.868) and Mistral-7B (0.868). The GCI dataset shows the largest relative improvement, increasing from 0.674 to 0.846 with GPT-4o. Although GPT-4o achieves the highest overall performance, which is likely because of its stronger reasoning and generation capability, all three LLMs substantially outperform the baseline when combined with RAFG. This improvement indicates that RAFG possesses good generalizability across LLMs of varying scales and architectures.

Information Gain. We further study how much addition information is introduced by the new features generated by RAFG. We adopt the *Information Gain* (IG) to measure the reduction in uncertainty of the target variable when the generated features are incorporated. A higher IG value indicates that

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TABLE II: Overall performance on various downstream tasks across different models. The best and second-best performance is set in **bold** and underline.

Metrics	Model	RF				DT				TT				TN			
Acc	Dataset	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA
	Raw	0.820	0.635	0.674	0.759	0.773	0.423	0.571	0.794	0.841	0.577	0.306	0.861	0.677	0.365	0.510	0.859
	Lasso	0.847	0.654	0.735	0.860	0.788	0.442	0.592	0.796	0.857	0.615	0.327	0.862	0.709	0.423	0.531	0.860
	RL	0.857	0.687	0.755	0.859	0.799	0.519	0.592	0.796	0.878	0.596	0.347	0.862	0.714	0.558	0.551	0.863
	CAAFE	0.862	0.673	0.776	0.863	0.820	0.577	0.612	0.797	0.884	0.654	0.408	0.863	0.741	0.577	0.592	0.865
	RAFG	**0.888**	**0.750**	**0.846**	**0.887**	**0.841**	**0.615**	**0.674**	**0.815**	**0.900**	**0.673**	**0.429**	**0.865**	**0.773**	**0.596**	**0.714**	**0.867**
Prec	Dataset	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA
	Raw	0.721	0.175	0.675	0.681	0.688	0.118	0.551	0.588	0.797	0.218	0.327	0.575	0.490	0.106	0.474	0.429
	Lasso	0.753	0.188	0.690	0.684	0.755	0.150	0.607	0.590	0.797	0.237	0.307	**0.599**	0.519	0.129	0.525	0.684
	RL	0.760	0.245	0.763	0.683	0.722	**0.319**	0.554	0.590	0.832	0.252	0.360	0.585	0.489	0.151	0.636	**0.806**
	CAAFE	0.753	0.237	0.768	0.724	**0.783**	0.173	0.607	0.592	0.832	0.321	0.409	0.572	0.536	0.196	0.601	0.715
	RAFG	**0.793**	**0.343**	**0.791**	**0.735**	0.777	0.261	**0.669**	**0.605**	**0.857**	**0.349**	**0.450**	0.570	**0.559**	**0.242**	**0.683**	0.723
Rec	Dataset	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA
	Raw	0.855	0.156	0.640	0.570	0.737	0.126	0.564	0.597	0.790	0.201	0.311	0.702	0.430	0.113	0.474	0.500
	Lasso	0.818	0.194	0.720	0.575	0.727	0.141	0.594	0.599	0.820	0.224	0.333	0.696	0.583	0.101	0.533	0.538
	RL	0.842	0.206	0.742	0.568	0.742	0.244	0.549	0.598	0.846	0.236	0.373	0.691	0.440	0.101	0.533	0.501
	CAAFE	0.890	0.337	0.766	0.559	0.763	0.262	0.603	**0.603**	0.832	**0.433**	0.310	0.705	0.653	0.116	0.592	0.575
	RAFG	**0.905**	**0.357**	**0.791**	**0.563**	**0.786**	**0.320**	**0.703**	0.596	**0.882**	0.353	**0.404**	**0.710**	**0.664**	**0.213**	**0.669**	**0.579**
F1	Dataset	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA	PDC	AID	GCI	DIA
	Raw	0.750	0.146	0.636	0.587	0.702	0.122	0.556	0.592	0.793	0.209	0.289	0.593	0.419	0.110	0.470	0.462
	Lasso	0.776	0.180	0.679	0.594	0.738	0.137	0.599	0.594	0.808	0.228	0.303	**0.622**	0.476	0.105	0.526	0.539
	RL	0.788	0.209	0.741	0.584	0.731	0.268	0.549	0.594	0.838	0.225	0.363	0.606	0.434	0.121	0.551	0.466
	CAAFE	0.790	0.243	0.758	0.572	0.772	0.188	0.601	0.597	0.832	**0.344**	0.350	0.589	0.508	0.137	0.579	0.595
	RAFG	**0.829**	**0.324**	**0.788**	**0.578**	**0.781**	**0.271**	**0.681**	**0.600**	**0.868**	0.341	**0.399**	0.587	**0.556**	**0.224**	**0.664**	**0.600**

TABLE III: Variation in the number of features.

Dataset	Original	RAFG	Dataset	Original	RAFG
PDC	756	770 (+14)	GCI	34	39 (+5)
AID	15	16 (+1)	DIA	21	23 (+2)

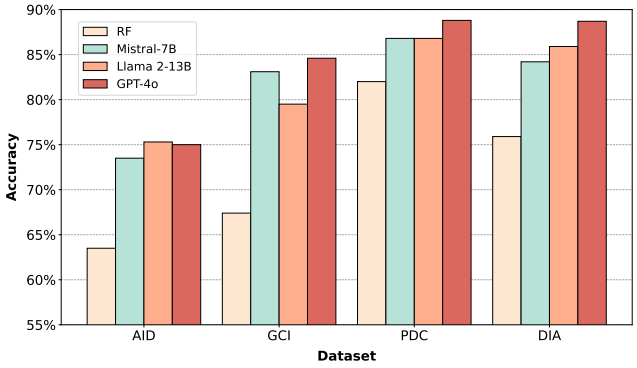


Fig. 3: Accuracy of using RF with RAFG and various LLMs.

the new feature provides novel, predictive information about the label's distribution not captured by the original feature set. As shown in Fig. 4, both the prediction accuracy and IG increase consistently with RAFG. In particular, for PDC and GCI, the IG increases over 200% and 400%. This result indicates that the RAFG-generated features bring new informative signals that make the target variable more predictable for the model. Further, by retrieving external context and generating transformed new features, RAFG scales the dimension of the feature space for the downstream classifiers to capture the latent structure behind the data.

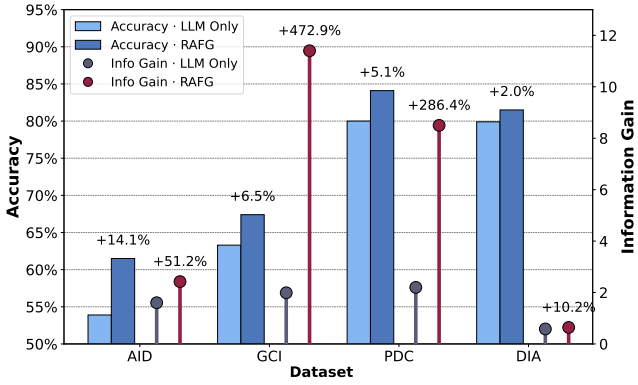


Fig. 4: Information gain with RAFG across different datasets.

Case Study. To further illustrate how RAFG generates meaningful and interpretable features, we conduct a detailed case study on the *Global Country Information Dataset 2023 (GCI)* [59] dataset. This dataset provides well-structured indicators, e.g., economic, demographic, and environmental attributes at the country level. We analyze how the new features contribute to downstream task performance by answering: *a)* what types of features does RAFG generate and their interpretability; *b)* how do these features quantitatively impact task performance and information gain; and *c)* how do these new features correlate with the original ones and whether they capture more domain-level relations.

a) Generated Features and Interpretability: In Table IV and V we list five new features generated by RAFG. Each feature includes a label, a calculation formula, and an auto-

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TABLE IV: Some case studies about new features generated by RAFG based on the GCI dataset.

Label	**Population Load Ratio**
Calculation	$\frac{\text{Population}}{\text{Land Area (Km}^2\text{)}}$
Reasoning	*Represents population density per square kilometer, indicating the population load of a country.*
Label	**Resource Utilization Rate**
Calculation	$\frac{\text{Agricultural Land (\%)} + \text{Forested Area (\%)}}{100}$
Reasoning	*Measures the proportion of land used for agriculture and forestry; higher values indicate greater resource use.*
Label	**Education Investment Effectiveness**
Calculation	$\frac{\text{Gross Primary Enrollment (\%)} + \text{Gross Tertiary Enrollment (\%)}}{2}$
Reasoning	*Reflects investment in primary and higher education, typically linked to better economic development.*
Label	**Environmental Stress Index**
Calculation	$\frac{\text{CO}_2\text{ Emissions}}{\left(\frac{\text{Forested Area (\%)}}{100} \times \text{Land Area (Km}^2\text{)}\right)}$
Reasoning	*Indicates environmental stress by measuring CO_2 emissions per unit of forest area. Higher values show greater pressure.*
Label	**GDP per Capita**
Calculation	$\frac{\text{GDP}}{\text{Population}}$
Reasoning	*Represents a country's economic level and standard of living.*

TABLE V: Comparison of applying RAFG on the GCI dataset. The blue highlighted text denotes the new descriptive information generated by the LLM.

Dataset	Number of Features	Data Description
Original	35	This comprehensive dataset provides a wealth of information about all countries worldwide, ..., enabling in-depth analyses and cross-country comparisons.
New	40 (+5)	This comprehensive dataset provides a wealth of information about all countries worldwide, ..., enabling in-depth analyses and cross-country comparisons. Newly added to this dataset are five key variables designed to deepen insights into economic pressures, population density, resource utilization, educational investments, and environmental by stress.

matically generated reasoning explanation. In this case, these features are mainly widely used socio-economic indicators, e.g., ***the Population Load Ratio*** [74] integrates ***Population*** and ***Land Area*** to describe population density, and ***Resource Utilization Rate*** [75] integrates ***Agricultural Land*** and ***Forested Area*** to describe ecological pressure. These clear and useful generations demonstrate that RAFG can maintain transparency and semantic meaning based on domain knowledge.

b) Performance and Information Gain Evolution: Fig. 5 illustrates how each new feature improves the classification accuracy and increases the dataset's information content. With five new features by RAFG, the accuracy improves by 10.2% and the IG increases from 1.99 bits to 11.4 bits. This trend demonstrates that each new feature provides distinct and useful information about the target variable.

c) Correlation: To verify that RAFG does not simply generate redundant transformations, we explore the correlation between the new features generated by RAFG and the existing features in the GCI dataset. Fig. 6 shows a strong positive correlation between these newly generated features and several key features in the dataset. For example, the ***Population Load Ratio (PLR)*** significantly correlated with geographic indicators like ***Environmental Stress Index***, which is because population density interacts with territorial constraints [76–78]. Similarly, ***Resource Utilization Rate (RUR)*** is strongly related to ***Population Load Ratio***. This correlation reflects a common real-world pattern in which denser populations cause greater pressure on environmental resources [79, 80]. In short, the correlations validate that the generated features capture

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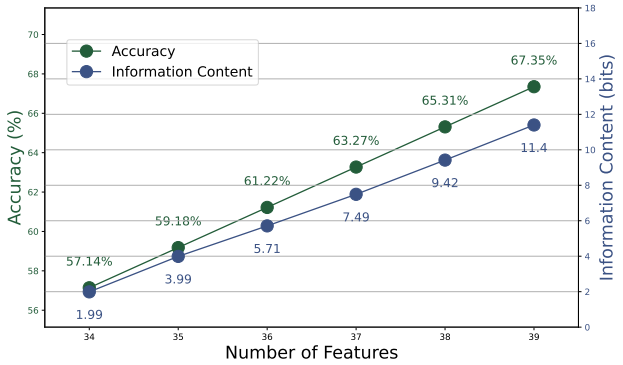


Fig. 5: Accuracy variation in RAFG feature generation process and the information gain for GCI dataset with DT model.

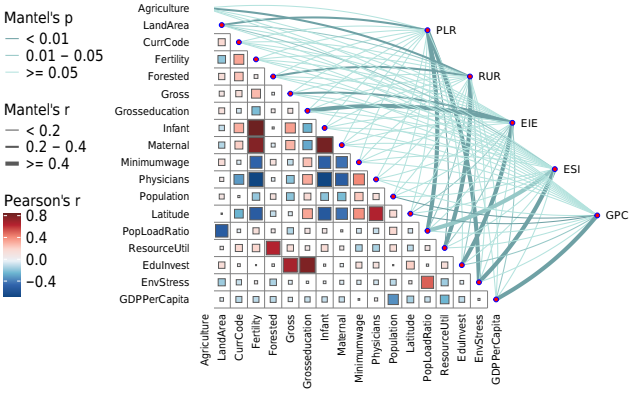


Fig. 6: Heatmap of the correlations between the newly generated features (right) and the original features (left).

higher-level interactions between the existing factors, which are implicit in the raw data.

VI. Conclusion

In this paper, we introduce a novel method, RAFG, that utilizes an LLM for retrieval-augmented feature generation. Our target is to enrich data by effectively utilizing the textual information from external knowledge and LLM's reasoning capability to generate domain-specific features. We adopt RAG technique to enhance the correctness and interpretability of feature generation. The generated features significantly enhance machine learning models without the need for high resource costs or domain-specific fine-tuning. The experimental results demonstrate that RAFG outperforms existing methods in generating useful features and enhancing performance across various domains. Besides, RAFG's automated and adaptable framework enable the consistent evolves with new data and shows great potential for broader adaptability in various fields.

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